

The Impact of Gaze and Hand Gesture Complexity on Gaze-Pinch Interaction Performances

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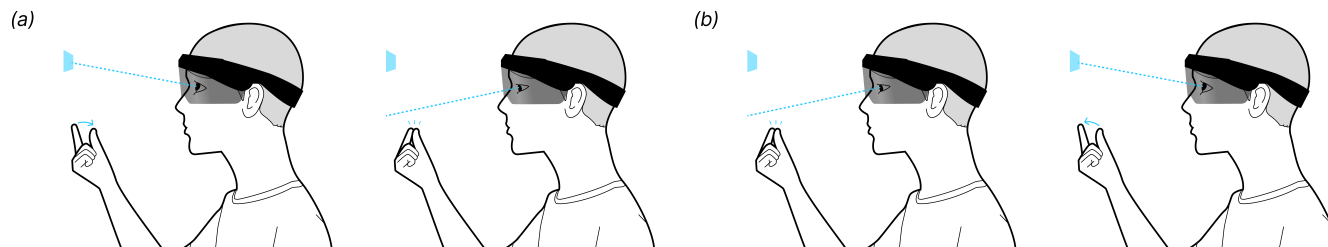


Figure 1: Gaze-pinch coordination errors during gaze-pinch interaction. A late trigger (a) occurs when the gaze moves before pinching, while an early trigger (b) happens when pinching precedes accurate gaze fixation.

ABSTRACT

This paper investigates user performance and gaze-hand coordination errors in gaze-pinch interaction across different task complexities involving gaze and hand gestures. We designed gaze-based single target pointing tasks with varying levels of gaze complexity: simple target, target with a visual cue, and target requiring visual search. Hand gesture complexity included simple thumb-index pinch and multi-finger pinching gestures (thumb-index, thumb-middle, thumb-ring). Our findings reveal that gaze-pinch coordination errors predominantly occur due to late triggering, where the gaze shifts away from the target before pinching begins. While hand gesture complexity did not significantly affect error rates, tasks requiring additional visual attention, like target recognition or visual search, increased gaze-pinch error rates. The study also highlights the impact of individual differences on gaze-hand coordination during interactive tasks. These individual differences underscore the need for personalized approaches in designing gaze-hand interaction systems that cater to diverse user profiles. Future work aims to address these challenges with larger participant groups and further explore factors influencing gaze-hand coordination in interactive systems.

CCS CONCEPTS

• **Human-centered computing** → **Gestural input**; *Virtual reality*; User studies.

KEYWORDS

Gaze-pinch interaction, Gaze-hand coordination, Input, Virtual Reality

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1 INTRODUCTION

Gaze-based interaction allows for faster [26], natural and intuitive pointing [25] in diverse interaction scenarios, such as desktop [6, 9], mobile [5, 13], and Head-Mounted Displays (HMDs) [23, 24]. Especially, commercial HMDs equipped with high-fidelity gaze tracking systems (e.g., Apple Vision Pro [10] and Meta Quest pro [18, 27]) have enabled more expressiveness with greater precision for interaction in Virtual Reality (VR) [3]. For example, gaze interaction has been coupled with multi-modal selection methods using hand actions [1, 4, 16, 17], speech [12, 21], and head movements [3] to trigger a target using the gaze-based cursor. For example, Lystbaek et al. [17] proposed the alignment between gaze and hand-pointing as input triggers in augmented reality. Among various gaze-hand interaction techniques, the gaze-pinch input technique [22] has gained prominence. This technique utilizes gaze as a pointer to focus on a target and pinch gestures detected by outward-facing cameras to trigger a selection. It has been increasingly adopted in recent commercial devices as the general input paradigm [11]. Gaze-pinch interaction is beneficial in real-world scenarios since it is robust to the Midas touch problem caused by unintended actions by the eyes [19, 22] and enables users to execute inputs with a simple touch between the fingers that can be detected requiring additional external equipment [15] or extensive body movement [7].



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This approach offers intuitive and advanced target selection without restrictions from the surrounding environment [17] and physical fatigue [22].

Despite the advantages of gaze-pinch interaction, previous works have noted that gaze-hand coordination problems, such as the situations in which the target is selected by manual command (e.g., pinch gesture) after the gaze moves to another target [2, 8, 14, 20] or before the gaze arrives at the target [14], limit the effectiveness of the technique [14]. Previous works noted that there is a time difference of around 100ms to 200ms between gaze and hand input during simple task performance [8, 20], and other variables such as the presence of hand haptics [20], and stress levels [8] are known to influence the timing of gaze and hand input. For example, Pfeuffer et al. [22] reported that timing errors from gaze-hand coordination could elevate input errors and interrupt users' experience in various input tasks.

In this work, to understand the pattern of gaze-hand coordination errors during various tasks with gaze-pinch input, we examined how user performance and the gaze-pinch coordination error data change as we vary task complexity in terms of both the visual (gaze) and manual (hand) task performed. For gaze complexity, we designed the gaze-based single target pointing task with three different visual stimuli: 1. target, 2. target + visual cue for next target, and 3. target requiring visual search. For hand gesture complexity, the target can be selected either by simple thumb-index pinch or one of three multi-finger pinching gestures: thumb-index, thumb-middle, and thumb-ring pinch. We illustrate details of the study design in Section 2.1. We investigated the differences in performance caused by these changes in a pilot study and discovered:

- **Gaze-pinch coordination error:** Tasks requiring more complex gaze behaviors, such as visual search or target recognition, increase the incidence of late triggering errors, where the gaze shifts away from the target before the pinch gesture occurs.
- **User performance data:** Tasks involving complex gaze behaviors or multi-finger gestures lead to higher error rates and longer task completion times due to increased cognitive and motor demands.

Additionally, our study suggested that individual differences may play an important role in handling the complexity of gaze and hand gestures. Understanding these individual differences is essential for designing adaptive interfaces for diverse user needs and preferences. Accordingly, These findings underscore the importance of considering task complexity and individual differences in designing effective gaze-hand interaction systems. They provide insights into optimizing interaction techniques to enhance usability, minimize errors, and improve user experience across different user profiles and interaction contexts.

2 STUDY

We conducted a study to gather gaze and hand gesture data from a commercial HMD to analyze patterns of gaze-hand coordination errors and user performances during gaze-pinch interaction. We implemented our study app on Meta Quest Pro, which is equipped with inward-facing eye sensors and an outward-facing camera to

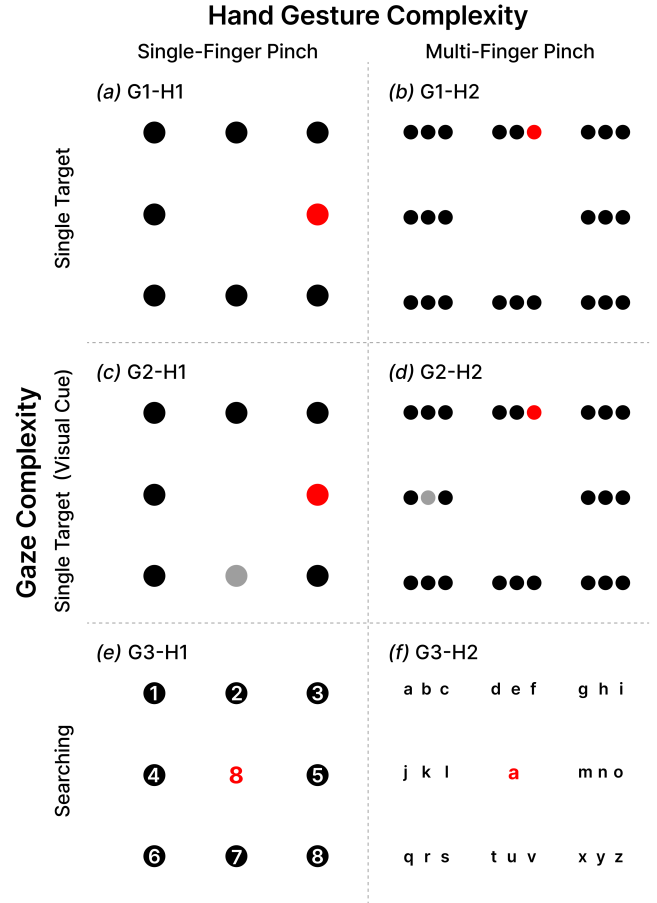


Figure 2: Task interfaces of our study for different conditions in terms of gaze and hand gesture complexity

capture gaze and pinch data. This study received approval from the local Institutional Review Board (IRB).

2.1 Participants

We recruited 6 participants (4 males, mean age of 24.83 (SD=2.32)) from local universities. 5 participants were right-handed, and 1 was ambidextrous. Among the participants, 4 had a right dominant eye, 1 had a left dominant eye, and 1 had no dominant eye. They were not familiar with VR devices (2.33/5 (SD=0.52)) and gaze interaction within VR devices (2.33/5 (SD=1.03)).

2.2 Design

In this study, there were eight target spaces (arranged in a three-by-three grid with the center point omitted, see Figure 2), and participants conducted a series of gaze for pointing and pinch using the right-hand for selection to perform sequential target selection. The study was designed around two key variables: gaze complexity and hand gesture complexity. The visual stimuli specifying the next target and required input action (e.g., single or multi-finger pinching) are designed to reflect these variables. The configurations for each variable are as follows:

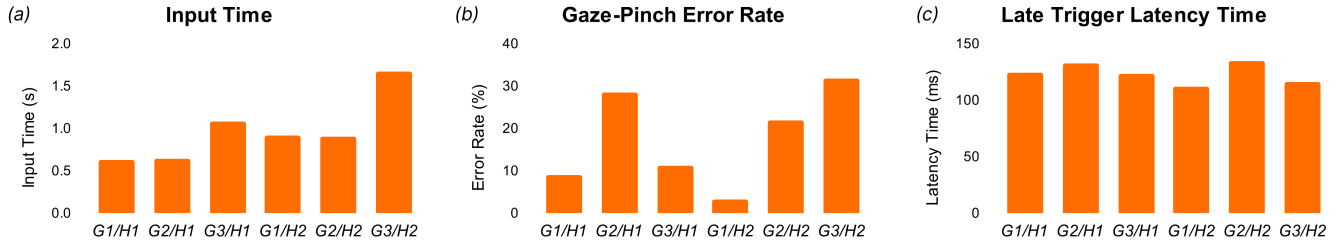


Figure 3: The average input time, gaze-pinch error rate, late trigger latency time for different task conditions

Gaze complexity: In the first gaze condition (G1), participants look at the red target to point to it (see the first row of Figure 2). In the second condition (G2), information about the next target is provided through a gray target. The experimental scene simultaneously displays the current target to be selected in red and the next target in gray (see the second row of Figure 2). Therefore, participants can gaze at the red target while being aware that they will be required to select the gray target next. Finally, in the third condition (G3), participants should search for a target based on a number in the center of the screen indicating (see the third row of Figure 2). Participants need to look at the number in the center of the scene first, then search and gaze at the target containing that number. After completing the input, the user needs to look at the center of the screen again to check the number indicating the next target.

Hand gesture complexity: In the first condition (H1), participants can make input simply using only a thumb-index pinch (see the first column of Figure 2). However, in the second condition (H2), participants need to select targets using multi-finger pinching gestures: thumb-index, thumb-middle, or thumb-ring pinches. For this condition, three targets are aligned in a row within each of the eight target spaces (for a total of 24 targets), and each of the three targets corresponds to one of three multi-finger pinching gestures. For example, the left target can be triggered by a thumb-ring pinching gesture, as the ring finger is spatially located on the left side of the right hand. Consequently, the left, middle, and right target in each target space is designed to be triggered by thumb-ring, thumb-middle, and thumb-ring pinching gestures.

2.3 Procedure

The experiment was conducted in an empty room where participants sat in a fixed posture in front of a desk. To prevent fatigue, participants put their elbows on the desk during the study. Participants first read the instructions and donned the Meta Quest Pro. For pinching gestures, they held their arms such that their hands faced toward their bodies and performed pinches with their right hand. Before the experiment began, participants adjusted their gaze using the calibration feature of the Quest Pro. Each task was performed after a short practice session with a maximum of five minutes. Each participant performed 96 sequential inputs (for H1: 8 target spaces by 12 repetitions, for H2: 8 target spaces by 3 fingers by 4 repetitions) in a session, and the target locations were randomly ordered. Participants were instructed to make their inputs as fast and accurately as possible. We conducted a total of 12 sessions

(three gaze complexities by two hand-gesture complexities by two repetitions) for each participant, and they were able to rest between each session. Additionally, after completing the two repetitions for each condition, participants reported workload on the NASA Task Load Index (TLX), a process that also enforced a short break.

3 RESULT

Of the 6912 completed trials, participants self-reported not pointing to the target with gaze (e.g., due to losing concentration) 6 times (0.09%). We excluded these 6 self-reported errors from data analysis due to their sparsity. Additionally, due to the low sample size ($N=6$), we opt to focus on describing the data characteristics and in-depth reports on individual performance in the subsequent analysis.

3.1 User Performance Data

Data for input time, measured from the time that a target appears until the pinch starts, from correctly completed trials is illustrated in Figure 3-(a). Not surprisingly, the input time increased as the hand gesture complexity increased. For example, the average time required for a single-finger pinch (H1) was 0.82 s (SD 0.27), but for a multi-finger pinch (H2), it increased to 1.16 s (SD 0.51). However, in terms of gaze complexity, the average input times indicate 0.77s (SD 0.27) for a single target (G1) and the same score of 0.77s (SD 0.23) for a target with the visual queue (G2), but it dramatically increased to 1.43 s (SD 0.41) for target requiring visual search (G3) due to additional time required for the searching task. We then examined the pinching finger errors and whether or not an appropriate finger was used for the pinching gesture during the multi-finger input condition (H2). The error rate increases slightly as gaze complexity

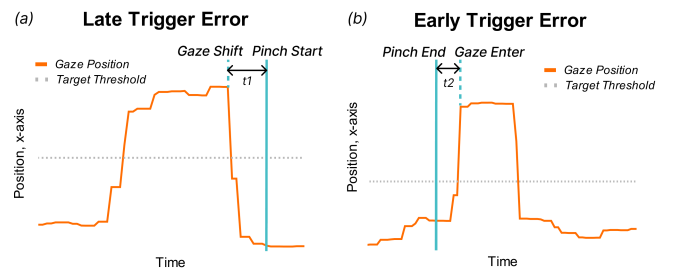


Figure 4: (a) Late trigger error, pinched after the gaze leaves the target space, (b) Early trigger error, pinched before the gaze enters the target space

	G1-H1		G2-H1		G3-H1		G1-H2		G2-H2		G3-H2	
	L.T	E.T	L.T	E.T	L.T	E.T	L.T	E.T	L.T	E.T	L.T	E.T
P1	N (#)	2	1	21	2	0	2	0	2	1	19	4
	t (ms)	35.0 (8.5)	42.0 (-)	113.7 (115.1)	56.0 (39.6)	-	209.0 (99.0)	-	132.0 (49.5)	28.0 (-)	320.0 (-)	271.5 (181.5)
P2	N (#)	5	2	97	14	19	23	5	2	126	2	160
	t (ms)	472.6 (256.3)	77.0 (89.1)	138.3 (110.8)	309.4 (203.6)	322.4 (222.9)	179.7 (197.5)	53.0 (18.6)	132.0 (9.9)	140.7 (73.5)	188.0 (127.3)	113.9 (64.7)
P3	N (#)	0	2	2	0	3	0	0	3	1	2	3
	t (ms)	-	28.5 (20.5)	28.0 (19.8)	-	22.0 (13.9)	-	-	69.7 (84.6)	28.0 (-)	63.0 (69.3)	46.0 (21.9)
P4	N (#)	67	5	149	6	36	6	19	4	109	4	129
	t (ms)	96.60 (96.29)	183.8 (217.3)	142.8 (92.2)	174.3 (214.2)	95.7 (136.6)	107.3 (124.6)	132.0 (129.2)	278.8 (277.6)	130.1 (107.0)	219.5 (140.3)	122.1 (108.2)
P5	N (#)	5	0	11	0	34	0	2	0	1	0	19
	t (ms)	125.0 (211.4)	-	32.2 (23.3)	-	55.7 (43.5)	-	63.5 (10.6)	-	42.0 (-)	-	72.6 (51.2)
P6	N (#)	5	8	18	5	4	1	0	0	1	3	4
	t (ms)	173.0 (180.3)	129.1 (128.0)	103.6 (134.9)	145.4 (200.2)	66.5 (42.6)	70.0 (-)	-	-	29.00 (-)	148.3 (197.3)	216.0 (118.3)

Table 1: The number of errors and late or early latency time for each participant under the different study conditions

increases as G1: 7.38% (SD = 4.74%), G2: 8.51% (SD = 4.88%), G3: 9.54% (SD = 3.64%).

3.2 Gaze-pinch Coordination Error

Gaze-pinch coordination error—whether or not the appropriate target was selected due to misalignment between gaze and pinch—can be classified into two distinct types: late trigger errors and early trigger errors (see Figure 4). Late trigger errors occur when gaze shifts from the target before the pinching begins, while early trigger errors happen when the pinch is initiated before the gaze has reached the target. For each error, late trigger latency time (t_1), from the end of the gaze pointing period until the pinch starts, and early trigger latency time (t_2), from the end of the pinch period until gaze pointing start, was measured.

The overall mean of gaze-pinch coordination error rate for each task is illustrated in Figure 3-(b). Regarding hand gesture complexity, the mean gaze-pinch error rate increases slightly with higher hand gesture complexity, from 16.06% (SD 22.44) in H1 to 18.92% (SD 29.03) in H2. Additionally, there are substantial increases in the mean gaze-pinch error rate for G2 (25.00%, SD 30.96) and G3 (21.44%, SD 27.89) compared to G1 (6.03%, SD 10.30). Interestingly, the majority of gaze-pinch errors were due to late triggers, accounting for 89.08% of all gaze-pinch errors. Due to the low occurrence rate and sample size of early triggers, we conducted further analysis only in terms of late trigger latency times. The mean late trigger latency time of 125.65 ms aligns with findings from prior studies [8, 20]. In terms of gaze complexity, G2 showed the largest average latency time at 140.39 ms (SD 174.85), compared to G1 at 120.86 ms (SD 153.25) and G3 at 117.72 ms (SD 176.23). For the hand gesture complexity, late trigger latency time decreases when hand gesture complexity increases in H1 with 137.13 ms (SD 193.80) compared to H2 with 123.13 ms (SD 162.64). However, due to the large variances in error rates and latency time, it is hard to report meaningful differences between different conditions.

To mitigate these large variances derived from variability within small samples (6 participants), we conducted individual-level analysis in terms of the number of errors and latency time (see Table 1). For example, in gaze-pinch coordination, while P3 rarely made errors (1.48%, SD=0.94) and P1, P5, and P6 also performed with generally low mean error rates of 5.38% (SD 6.12). On the other hand, P2 and P4 showed distinctly large mean error rates of 43.66%

(SD 30.32). This represents more than 80% of gaze-pinch errors that occurred over all participants. In this result, most general gaze-pinch error trends (in terms of gaze and hand-gesture complexity) are derived from these two participants, P2 and P4. Notably, P4 had a markedly high error rate of 37.5% even in the simplest condition (G1-H1), where others made few errors (1.8%). Despite the small sample size, this analysis highlights both the complexity of gaze and hand gestures and that individual differences strongly impact gaze-hand coordination errors.

4 DISCUSSION AND CONCLUSION

This work explored the impact of gaze and hand gesture complexity on user performance and gaze-hand coordination errors during gaze-pinch interaction. Our findings revealed that gaze-pinch coordination errors occurred predominantly due to late triggering. Errors increased in tasks requiring additional visual attention, such as recognizing the next target or searching for the target's location. In contrast, hand gesture complexity did not significantly affect the error rates. Additionally, we observed individual differences in gaze-hand coordination errors, which indicate that users with different levels of experience or cognitive workload may exhibit varying degrees of accuracy in gaze targeting and hand gestures. Understanding these individual differences is essential for developing adaptive interfaces that can accommodate user diversity.

However, this study has several limitations. The most significant limitation is the small sample size of six participants, which is not sufficient to represent general gestural characteristics during gaze-pinch interaction. Future research will involve a larger participant group to obtain more generalizable results. Additionally, individual differences in making gaze-pinch errors were observed. Conducting an in-depth analysis with a larger sample size would help identify the specific characteristics and underlying reasons for these errors, providing more comprehensive insights into gaze-pinch coordination.

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